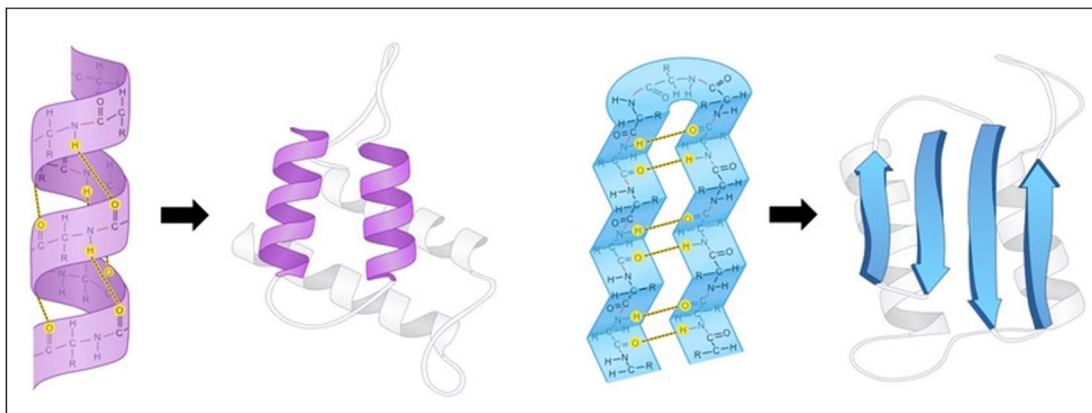


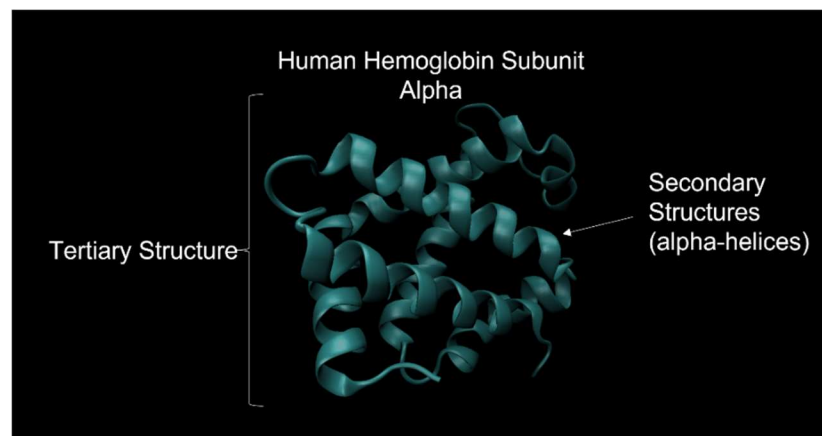
The Four Levels of Protein Structure

Protein structure is a broad term that encapsulates four different levels of description. A protein's **primary structure** refers to the amino acid sequence of the polypeptide chain. The primary structure of human hemoglobin subunit alpha can be downloaded [here](#), and the primary structure of the SARS-CoV-2 spike protein can be downloaded [here](#).

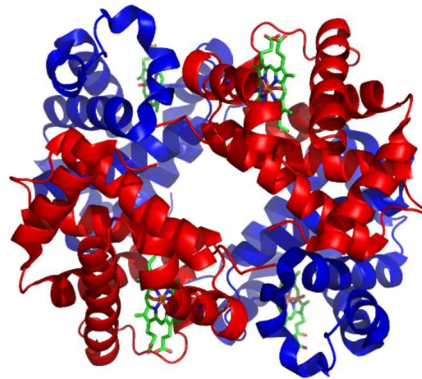
A protein's **secondary structure** describes its highly regular, repeating intermediate substructures that form before the overall protein folding process completes. The two most common such substructures, shown in the figure below, are **alpha helices** and **beta sheets**. Alpha helices occur when nearby amino acids wrap around to form a tube structure; beta sheets occur when nearby amino acids line up side-by-side.



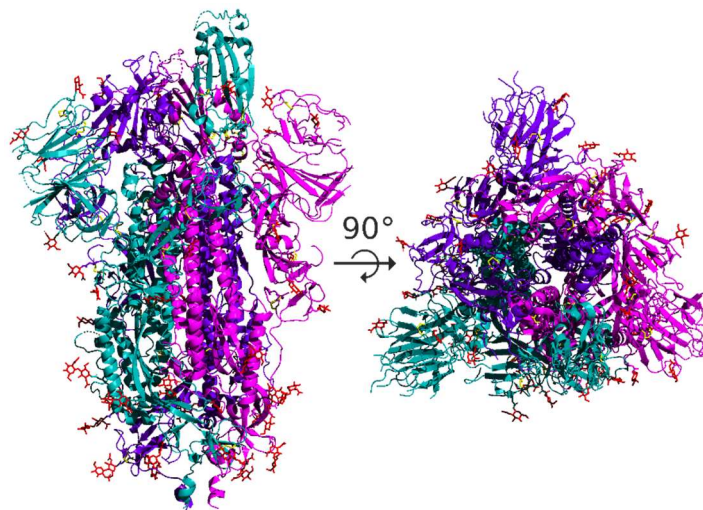
A protein's **tertiary structure** describes its final 3D shape after the polypeptide chain has folded and is stable. Throughout this module, when discussing the “shape” or “structure” of a protein, we are almost exclusively referring to its tertiary structure. The figure below shows the tertiary structure of human hemoglobin subunit alpha. For the sake of simplicity, this figure does not show the position of every atom in the protein but rather represents the protein shape as a composition of secondary structures.



Finally, some proteins have a **quaternary structure**, which describes the protein's interaction with other copies of itself to form a single functional unit, or a **multimer**. Many proteins do not have a quaternary structure and function as an independent monomer. The figure below shows the quaternary structure of hemoglobin, which is a multimer consisting of two alpha subunits and two beta subunits.



As for SARS-CoV and SARS-CoV-2, the spike protein is a **homotrimer**, meaning that it is formed of three essentially identical units called **chains**, each one translated from the corresponding region of the coronavirus's genome; these chains are colored differently in the figure below. When discussing the structure of the spike protein, we typically are referring to the structure of a single chain.



The structural units making up proteins are often hierarchical, and the spike protein is no exception. Each spike protein chain is a **dimer**, consisting of two subunits called **S1** and **S2**. Each of these subunits further divides into **protein domains**, distinct

structural units within the protein that fold independently and are typically responsible for a specific interaction or function. For example, the SARS-CoV-2 spike protein has a **receptor binding domain (RBD)** located on the S1 subunit of the spike protein that is responsible for interacting with the human ACE2 enzyme; the rest of the protein does not come into contact with ACE2. We will say more about the RBD soon.

Proteins seek the lowest energy conformation

Now that we know a bit more about how protein structure is defined, we will discuss why proteins fold in the same way every time. In other words, what are the factors driving the magic algorithm?

Amino acids' side chain variety causes amino acids to have different chemical properties, which can lead to different conformations being more chemically “preferable” than others. For example, the table below groups the twenty amino acids commonly occurring in proteins according to chemical properties. Nine of these amino acids are **hydrophobic** (also called **nonpolar**), meaning that their side chains tend to be repelled by water, and so we tend to find these amino acids sheltered from the environment on the interior of the protein.

AMINO ACID			
Nonpolar, aliphatic R groups			
	Glycine	Alanine	Valine
	Leucine	Methionine	Isoleucine
	Serine	Threonine	Cysteine
Polar, uncharged R groups			
	Proline	Asparagine	Glutamine
AMINO ACID			
Positively charged R groups			
	Lysine	Arginine	Histidine
Negatively charged R groups			
	Aspartate	Glutamate	
Nonpolar, aromatic R groups			
	Phenylalanine	Tyrosine	Tryptophan

We can therefore view protein folding as finding the tertiary structure that is the most *stable* given a polypeptide's primary structure. Any system of chemical reactions tends to move toward equilibrium, and the same principle is true of protein folding; when a protein folds into its final structure, it is obtaining a conformation that is as chemically stable as possible.

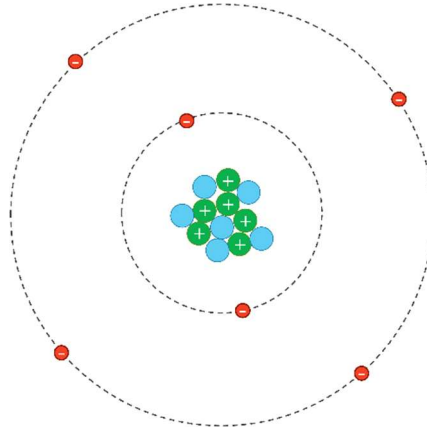
To be more precise, the **potential energy** (sometimes called **free energy**) of a molecule is the energy stored within an object due to its position, state, and arrangement. In molecular mechanics, the potential energy is made up of the sum of **bonded energy** and **non-bonded energy**.

As the protein bends and twists into a stable conformation, bonded energy derives from the protein's covalent bonds, as well as the bond angles within and between adjacent amino acids.

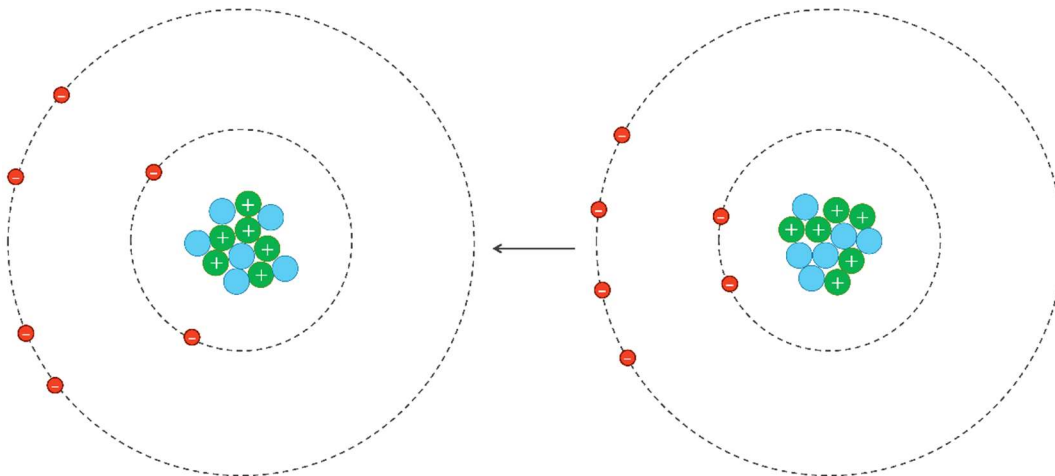
Non-bonded energy comprises **electrostatic interactions** and **van der Waals interactions**. Electrostatic interactions refer to the attraction and repulsion force from the electric charge between pairs of charged amino acids. Two of the twenty standard amino acids (arginine and lysine) are positively charged, and two (aspartic acid and glutamic acid)

are negatively charged. Two nearby amino acids of opposite charge may interact to form a **salt bridge**. Conformations that contain salt bridges and keep apart similarly charged amino acids will therefore have lower free energy contributed by electrostatic interactions.

As for van der Waals interactions, atoms are dynamic systems, with electrons constantly buzzing around the nucleus, as shown in the figure below.



However, due to random chance, the electrons in an atom could momentarily be unevenly distributed on one side of the nucleus. Because electrons are negatively charged, this uneven distribution will cause the atom to have a temporary negative charge on the side with the excess electrons and a temporary positive charge on the opposite side. As a result of this charge, one side of the atom may attract only the oppositely charged components of another atom, creating an **induced dipole** in that atom in turn as shown in the figure below. Van der Waals forces refer to the attraction and repulsion between atoms because of induced dipoles.



As the protein folds, it seeks a conformation of *lowest* total potential energy based on the combination of all these forces. For an analogy, imagine a ball on a slope, as shown in the following figure. The ball will tend to move down the slope unless it is pushed uphill by some outside force, making it unlikely that the ball will wind up at the top of a hill. We will keep this analogy in mind as we return to the problem of protein structure prediction.

